

Shuocheng Guo, Ph.D.

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Summary — Shuocheng holds a **Ph.D. in Transportation Systems Engineering** from **The University of Alabama (UA)**, with expertise in **Operations Research (OR)** and **Data Science (DS)**. He is currently working as a **Research Scientist at Amazon**, aiming to leverage his strong optimization skills to tackle real-world challenges.

Technical Skills

Languages Python (numpy&pandas), R, Julia, SQL
Data Science AWS, Power BI, Node.js, deck.gl
Optimization XPRESS, Gurobi, MOSEL
Operations Research VRP, MILP, Warehouse, Assignment

Experience

Amazon, POD Science

Research Scientist & Applied Scientist

San Diego, CA

Feb 2025 – Present

Optimizing Print-on-Demand Service (Key Skills: Python, XPRESS, OR, DS, AWS)

- Worked on optimizing the manufacturing system to support POD business, leading to \$X M cost savings.

Amazon, Amazon Fulfillment Technologies (AFT)

Research Scientist Intern

Bellevue, WA

Jun 2024 – Sep 2024

Optimizing Order Fulfillment in Amazon Bulk Business (Key Skills: Python, XPRESS, OR, AWS)

- Developed a **mathematical optimization** model to solve order picking problems in bulk businesses, significantly **improving space utilization** and **operational efficiency**.
- Implemented two **acceleration techniques** to enhance computational performance, improving CPU time by > 90% while maintaining optimality gaps < 10% and reducing > 90% of MIP variables.
- Designed an automated data pipeline using **SQL and AWS** to streamline data retrieval for optimization.

Delta Air Lines, Operations Decision & Data Science (ODDS)

Ph.D. Intern – Decision Scientist

Atlanta, GA

Jan 2024 – May 2024

Optimizing Haul Service Baggage Handling at ATL Airport (Key Skills: Python, Gurobi, OR, DS, AWS, Git)

- Developed and refined mathematical models and heuristics to improve baggage handling efficiency, achieving a **20% reduction in CPU times** and enabling faster transition to production.
- Conducted **sensitivity analyses** to evaluate varying constraints and objectives, improving operational efficiency.
- Enhanced the user interface platform's pipeline using **Power BI, SQL, and Python**, streamlining data processing.

The University of Alabama, Alabama Transportation Institute

Graduate Research Assistant

Tuscaloosa, AL

Aug 2020 – Dec 2024

Route Scheduler for Para-Transit System (Skills: Python, Julia, Node.js, OR, GNN, Gurobi)

- Built and deployed a **Column Generation toolkit** in **Julia/Python**, reducing COVID-19 exposure risk by **60%** and detours by **30%**, implemented in Birmingham, AL.
- Proposed ML-based acceleration via **Graph Neural Networks (GNN)**, achieving a **40% reduction** in CPU times with < 10% optimality gap.

Agent-Based Simulation Platform for Electric Taxi (Skills: Python, Simulation, Deep Reinforcement Learning)

- Built an **Electric Shared Mobility Simulator** in **Python** for hybrid taxi fleets.
- Developed a **Deep Reinforcement Learning (DRL)** framework for taxi relocation using **PyTorch**, increasing hourly revenue by **14.8%** and fulfilled orders by **2.7%** compared to baseline DQN.

Education

The University of Alabama

Ph.D. in Transportation Systems Engineering

Tuscaloosa, AL

Sep 2020 – Dec 2024

- Award:** Outstanding Ph.D. Researcher at CCEE Dept. (1/20); Finalist, Transit Data Challenge - TRBAM 2023 (4/25)

University of Illinois at Urbana-Champaign

M.Sc. in Transportation Systems Engineering

Urbana, IL

Sep 2018 – May 2020

Central South University

B.Sc. in Civil Engineering

Changsha, China

Sep 2015 – Jun 2018

Publications

Short Summary: I have 10+ strong publications and manuscripts related to OR, DS, and ML, published in *Transportation Research Part A/B/C/D* and *IEEE Transactions on ITS*. Additionally, I have presented at 5 peer-reviewed conferences and over 20 oral and poster sessions, including TRBAM 2021–2025 and INFORMS 2022–2024. See my Google Scholar for full details.